



REAL-TIME CHAT,
SIMPLIFIED.

Yoo Chat

PROJECT DOCUMENTATION

A cross-platform mobile chat application for Android and iOS, built for secure profiles, contact management and instant messaging with sent, delivered and read indicators.

01

CROSS-PLATFORM

Android & iOS

02

REAL-TIME CHAT

WebSocket messaging

03

SECURE BACKEND

Java & MySQL

React Native

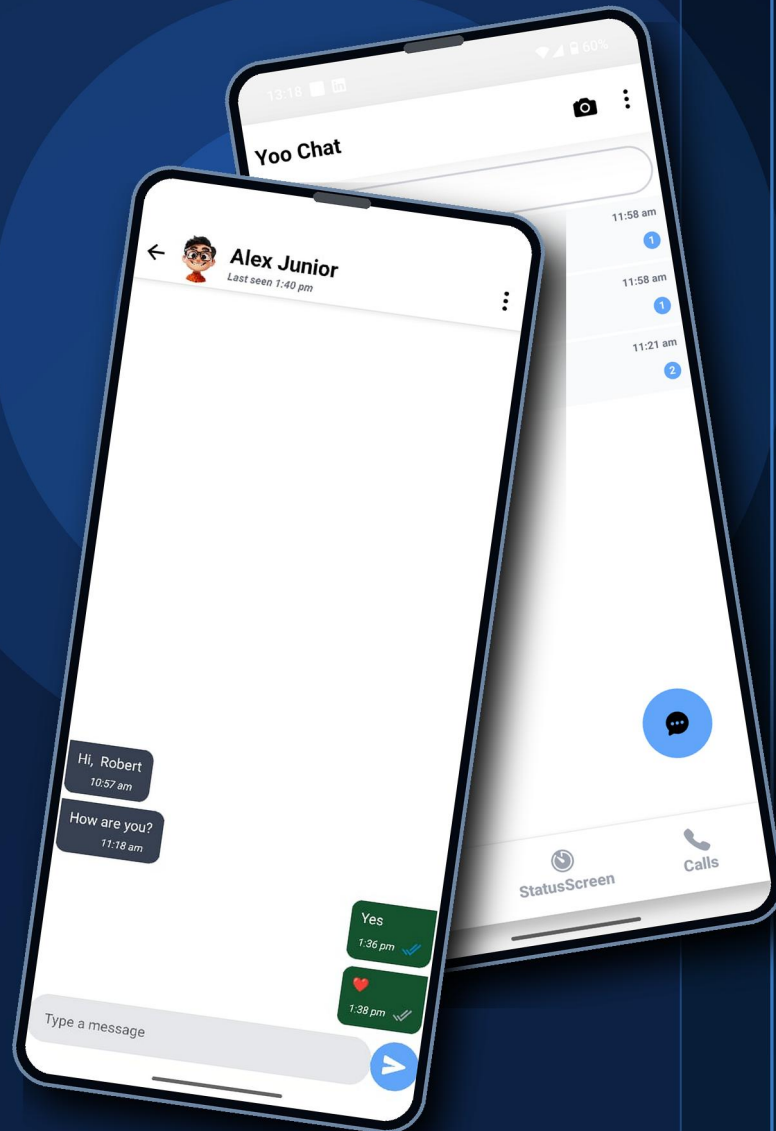
TypeScript

Java Servlets

WebSocket

Hibernate

MySQL



CONNECTED. DELIVERED.

Real-time messaging experience

PREPARED BY

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Software Engineering Project

PROJECT

Cross-Platform Mobile Application

React Native • Java EE • WebSocket • MySQL

Acknowledgement

I would like to express my gratitude to all those who supported me to build this Yoo Chat app successfully and who guided me throughout development. I would like to specially thanks to my lecturers for their greatest guidance and knowledge. Also my friends who supported me and my family for their encouragement. Their support made it possible to complete Yoo Chat app successfully.

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Project Overview

Yoo Chat is a cross platform mobile chat application that developed for Android and iOS using React Native. Yoo Chat App allows users to communicate through real time. It includes user friendly modern interface. The Yoo Chat App supports user authentication, contact management, profile customization, real time messaging with sent, delivered and seen indicators. The app uses java servlet for backend and it running on glassfish server. It uses WebSocket for real time interactions of users. This project implemented as a learning project.

Objectives

- To develop real time communication mobile application.

The primary goal of the Yoo Chat Application is build real time functional chat application for users that supports to send and receive messages instantly. Its similar to WhatsApp. This real time data transmission is done by using WebSocket technology to ensure smooth and clear conversations.

- To implement secure and reliable user authentication.

This application prevents unauthorized access and maintains the privacy of user details and chats.

- To provide user friendly smooth chat experience.

Yoo Chat application focus on very simple and user friendly interface that easy to navigate. Specially message indicators (sent, delivered and read) and profile customization.

- To support both Android and iOS platforms.

By using React Native the app developed as once and deployed for both Android and iOS platforms.

- To explore practical integration of React Native, Java Servlets, WebSockets, and MySQL.

The major learning object is use multiple technologies effectively like React Native for frontend, Java Servlet for backend and MySQL for database. Also WebSocket used for real time conversations.

- To implement features as profile management, contact addition and message delivery status.

Yoo Chat Application focus on provide all the main functionalities of a chat app including sign up, adding contacts, updating profile details, real time chatting and indicators for messages (sent, delivered, read).

Technology Stack

- Frontend : Used React Native and typescript for the frontend because the project should support to both Android and iOS.
- UI Styling : NativeWind for the UI Styling.
- Backend : Used javax.servlet for handle API requests and hibernate object relational mapping for the database.
- Real time communication : Used websocket for the real time data transferring and fetch API for REST calls.
- Database : Used MySQL for storing user, friend list and chat data.
- Hosting : Used glassfish server during development and made accessible to mobile using ngrok for tunneling.

System Architecture

- Client Layer : The client layer of the Yoo Chat App is a cross platform mobile application that used React Native for build. It is supporting to both Android and iOS smoothly.
- Server Layer : The server layer implemented using Java Servlets with Hibernate for handling API request and database operations.
- Communication Layer : The communication layer powered by websocket to ensure smooth communication by instant data transfer between client and server.
- Database Layer : The database layer uses MySQL to store user data, Friend lists and chat data. During the development, the backend hosted on glassfish server and enable to the mobile by using ngrok.

Features

- User sign up and sign out
- Add and update profile picture
- Add new contacts
- Real time chatting
- Last seen status
- Dark mode, light mode and system theme mode
- Sent delivered and read ticks on messages
- Emoji support for the messages
- Alert messages for new message

User Interface (UI) Design

Splash Screen

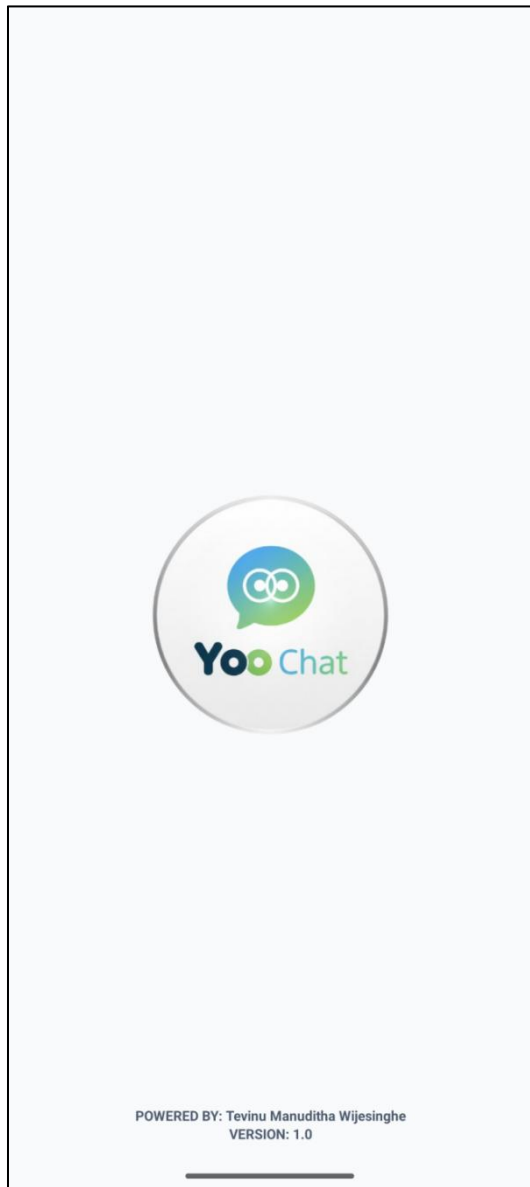


Figure 1 : Splash Scree Light Theme

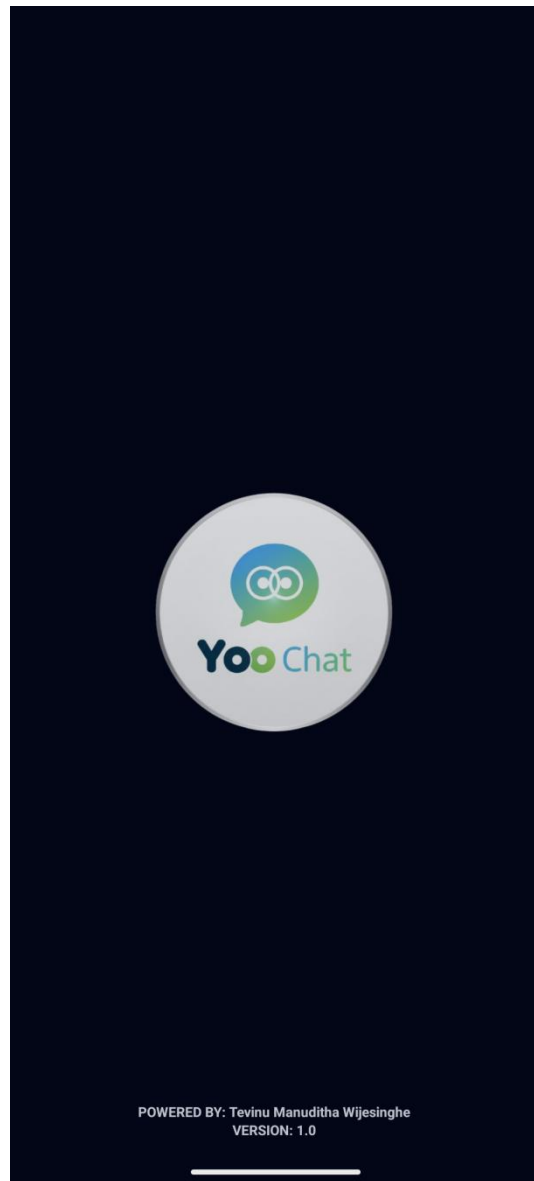
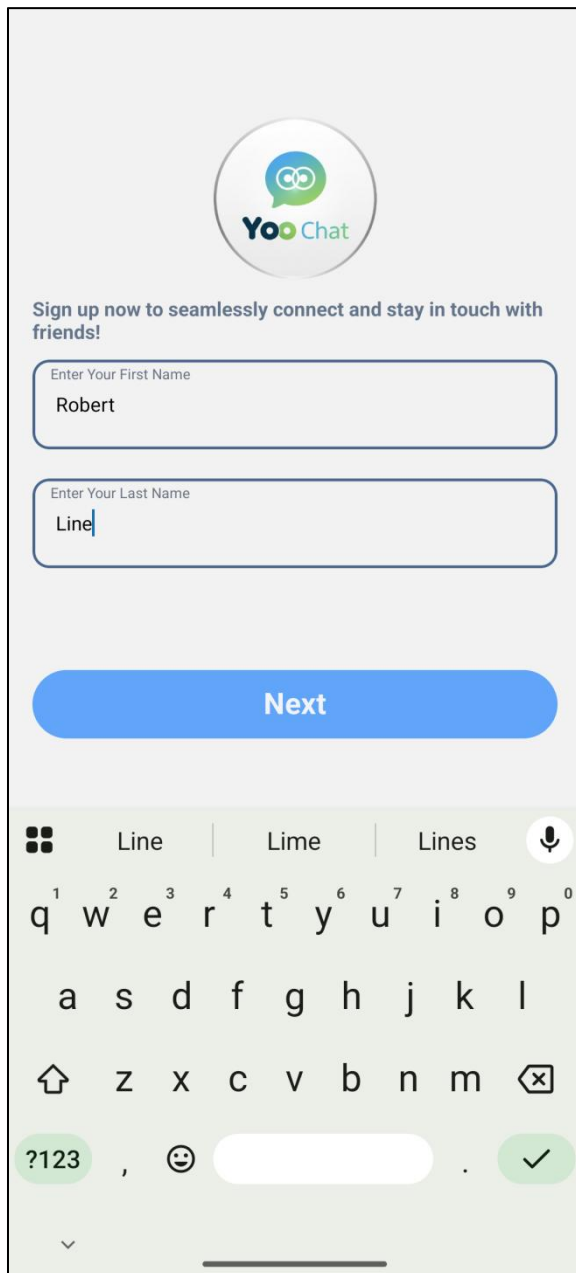


Figure 2 : Splash Screen Dark Theme

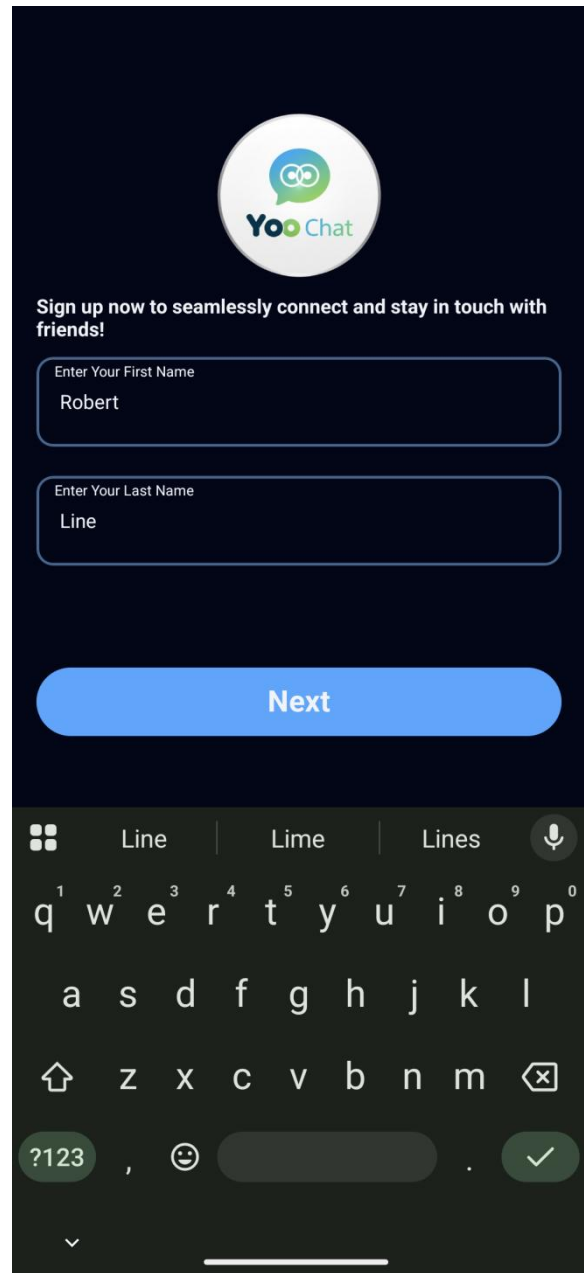
This is the Splash Screen of the Yoo Chat App. The Application starts with this screen.

Sign Up Screen



The Sign Up Screen in light theme features a light gray background. At the top center is the YooChat logo, which consists of two overlapping speech bubbles (one blue, one green) with the text "YooChat" below them. Below the logo is the text "Sign up now to seamlessly connect and stay in touch with friends!". There are two text input fields: the first is labeled "Enter Your First Name" and contains the text "Robert"; the second is labeled "Enter Your Last Name" and contains the text "Line". Below the input fields is a large blue button with the text "Next". At the bottom of the screen is a standard iOS-style keyboard with a light gray background and white text.

Figure 3 : Sign Up Screen Light



The Sign Up Screen in dark theme features a dark blue background. At the top center is the YooChat logo, which consists of two overlapping speech bubbles (one blue, one green) with the text "YooChat" below them. Below the logo is the text "Sign up now to seamlessly connect and stay in touch with friends!". There are two text input fields: the first is labeled "Enter Your First Name" and contains the text "Robert"; the second is labeled "Enter Your Last Name" and contains the text "Line". Below the input fields is a large blue button with the text "Next". At the bottom of the screen is a standard iOS-style keyboard with a dark gray background and white text.

Figure 4 : Sign Up Screen Dark

This is the Sign Up Screen of the Yoo Chat App. After the splash screen user can see this screen and user must enter first name and last name for move next page.

Contact Screen

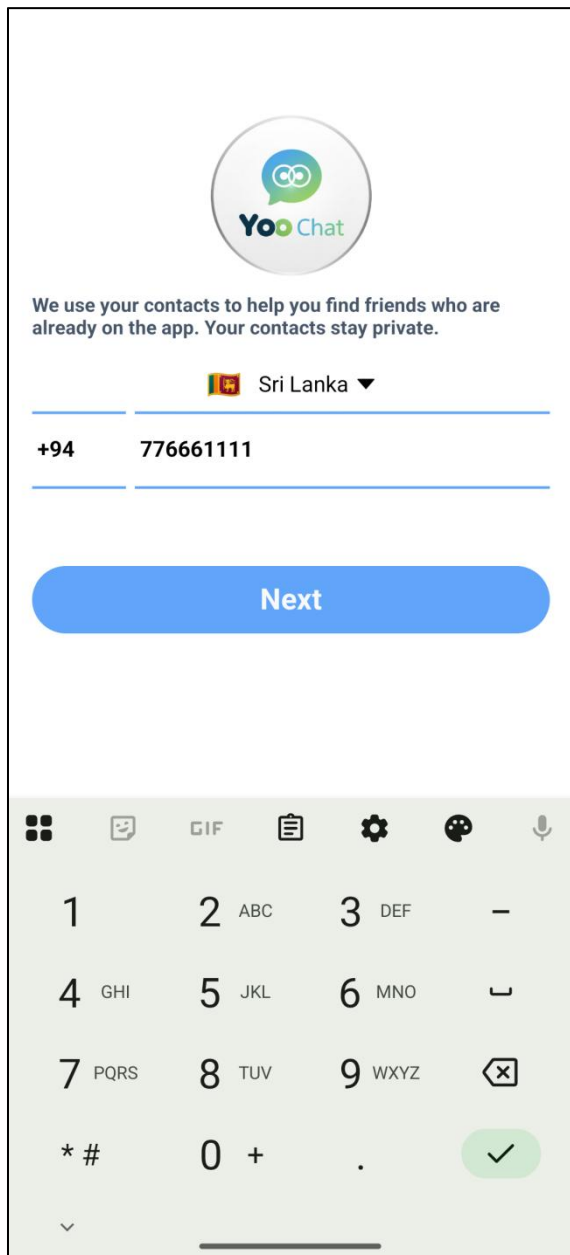


Figure 5 : Contact Screen Light Theme

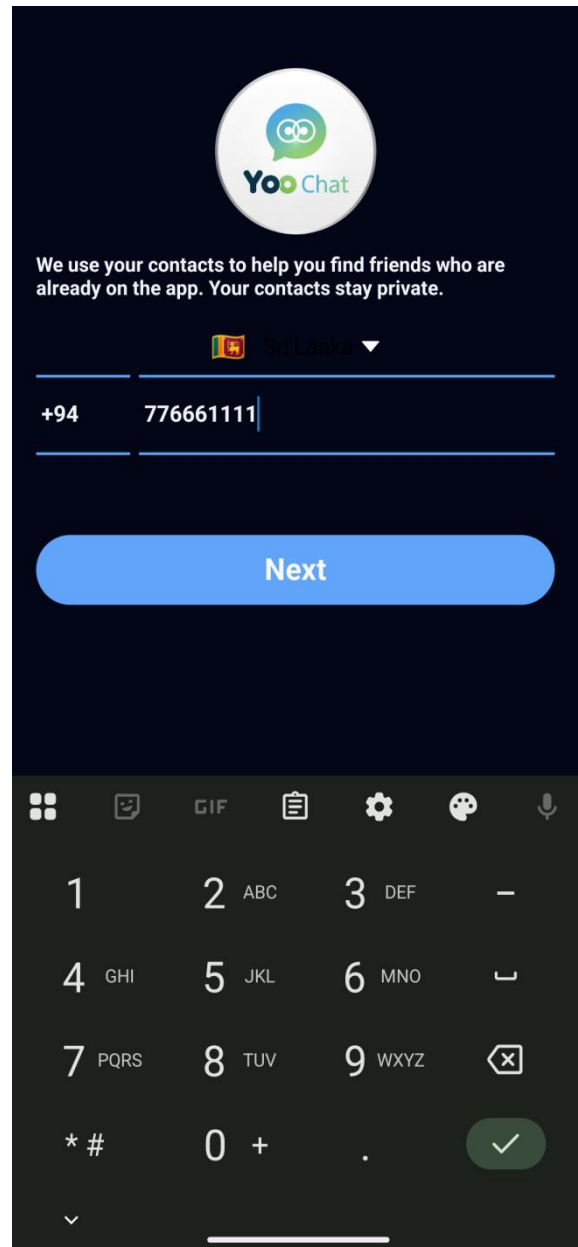


Figure 6 : Contact Screen Dark Theme.

This is the Contact Screen of the Yoo Chat App. After the sign up first screen user can see this screen and user must enter country code and Phone Number for move next page.

.Avatar Screen

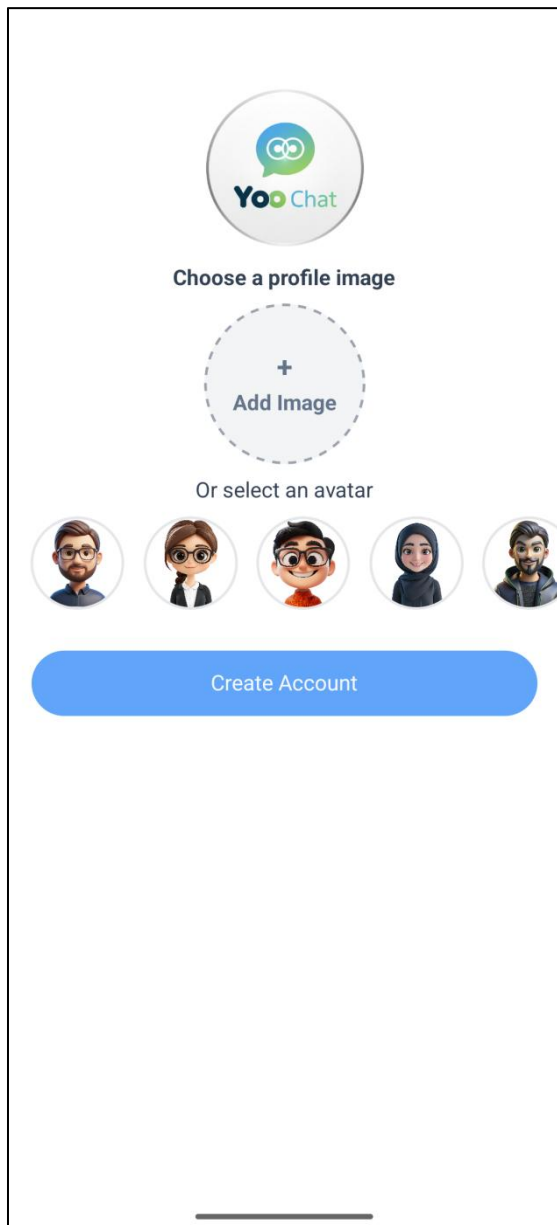


Figure 7 : Avatar Screen Light Theme

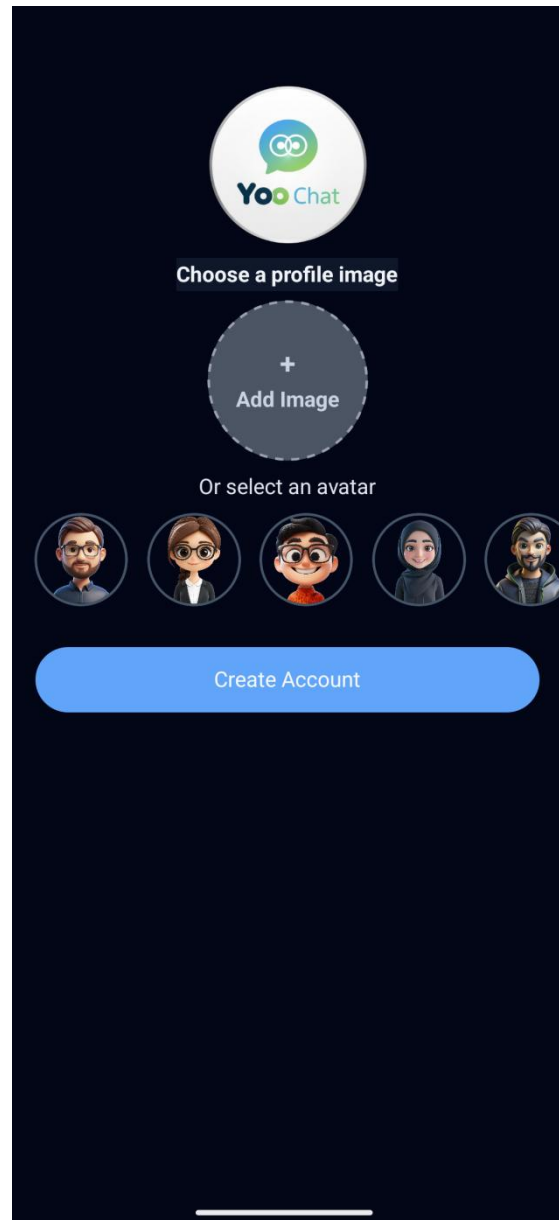


Figure 8 : Avatar Screen Dark Theme

This is the Avatar Screen of the Yoo Chat App. After the entering contact number, user can see this screen and user must select a profile picture to create account.

Home Screen

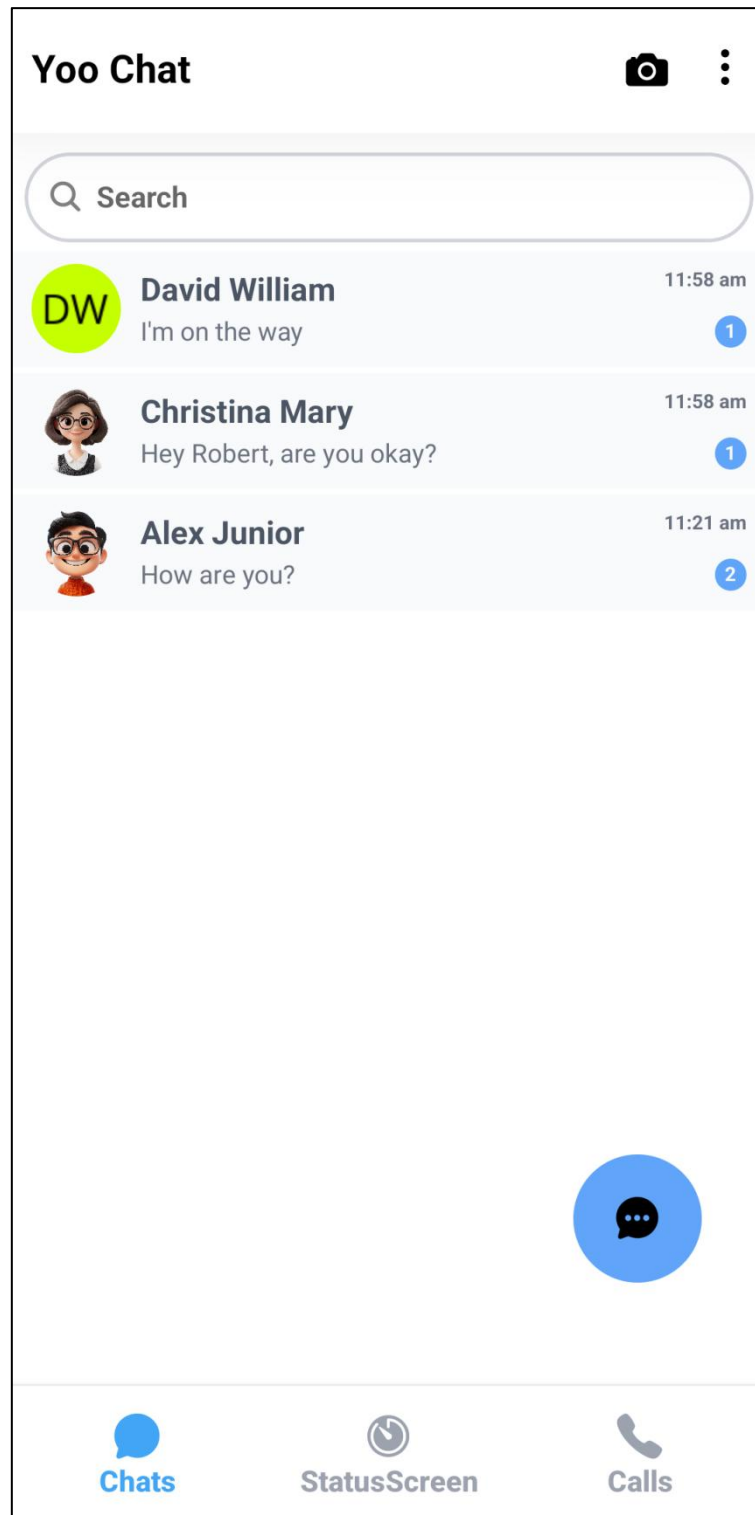


Figure 9 : Home Screen Light Theme

This is the HomeScreen of the Yoo Chat App. After the sign up process, user can see this screen and it lists the recent chats. User can touch a chat and read the messages and reply.

Singe Chat View

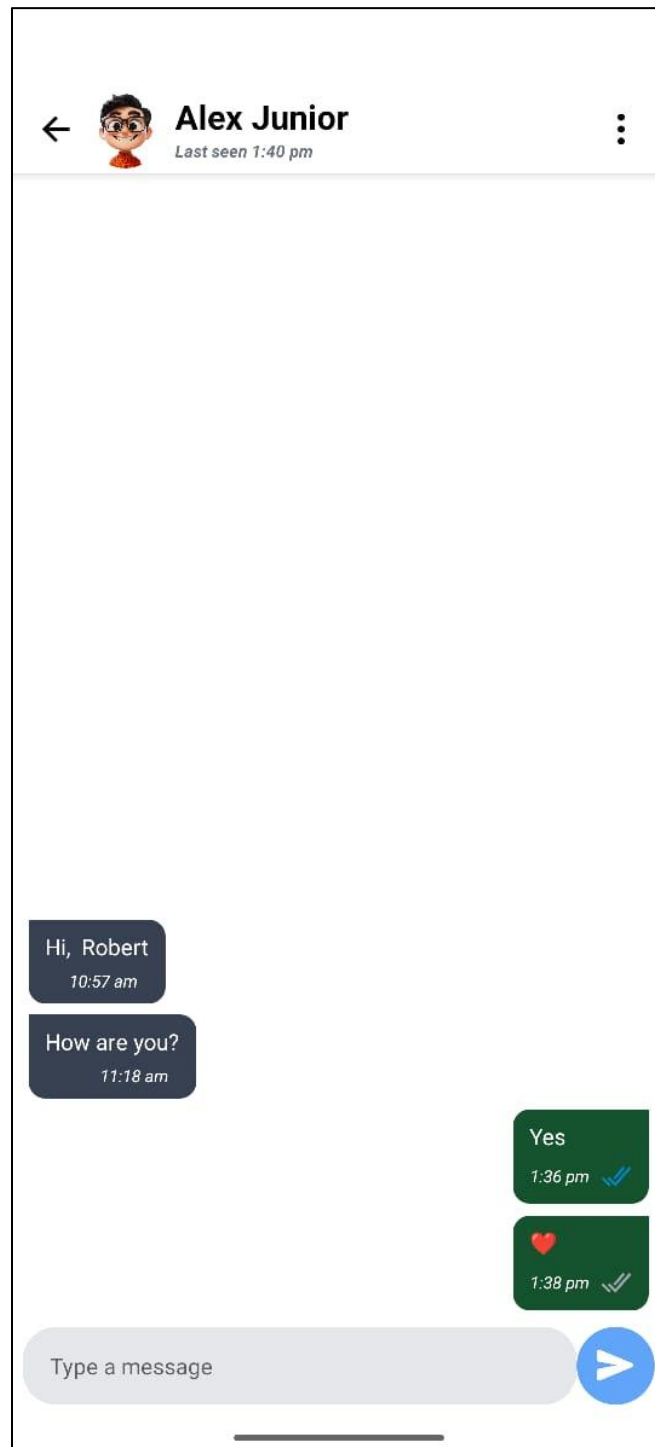


Figure 10 : Single Chat Light Theme

This is the Single chat screen of the Yoo Chat App. With this screen user can read the messages, reply and see friend last seen.

Options

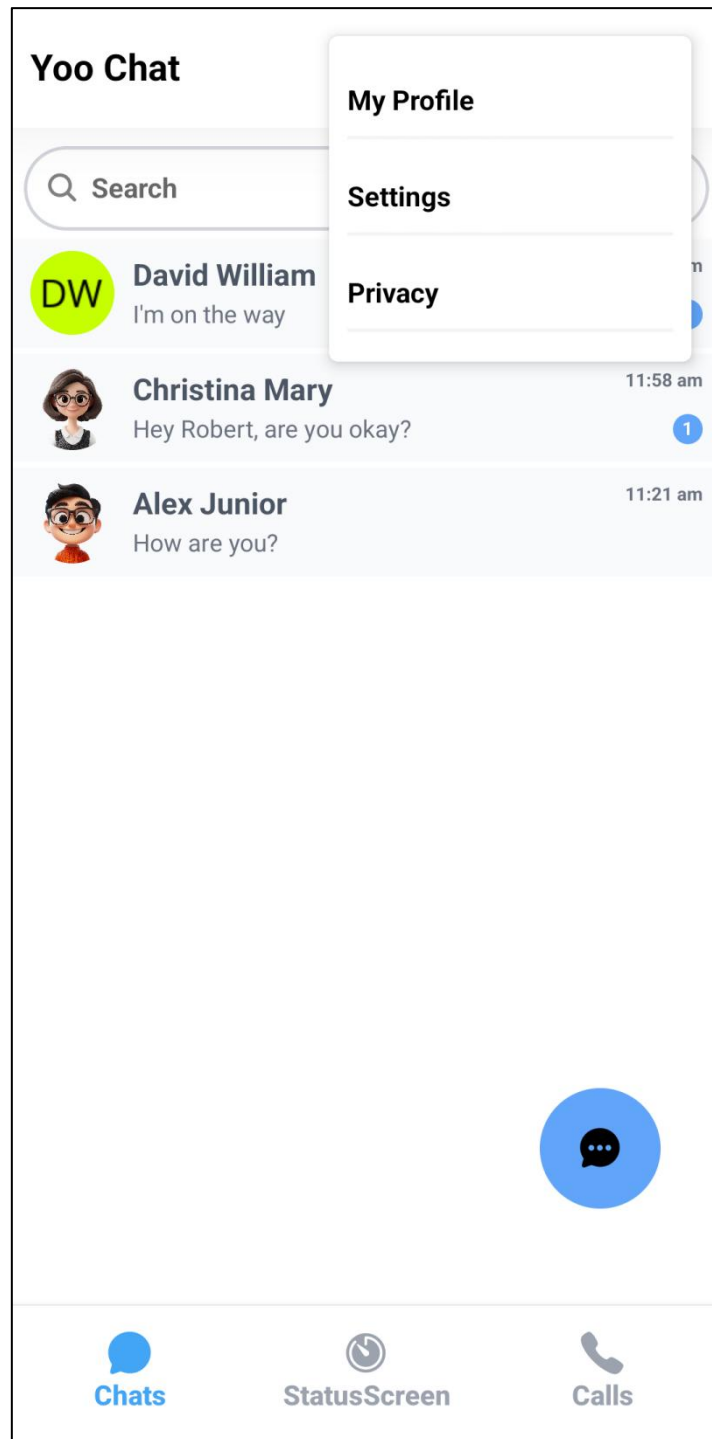


Figure 11 : Options

This is the Options of the Yoo Chat App. User can select options.

My Profile Screen

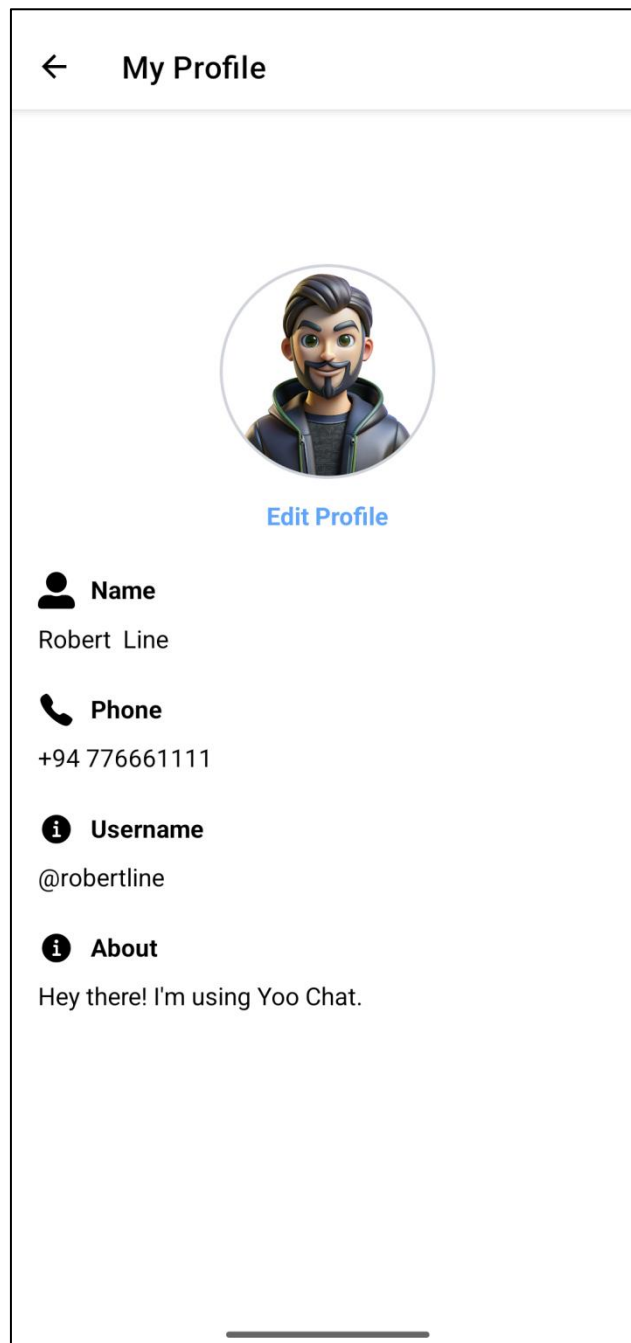


Figure 12 : Profile Screen Light Theme

This is profile screen of Yoo Chat App. User can see their profile details and user can update profile picture.

Settings Screen

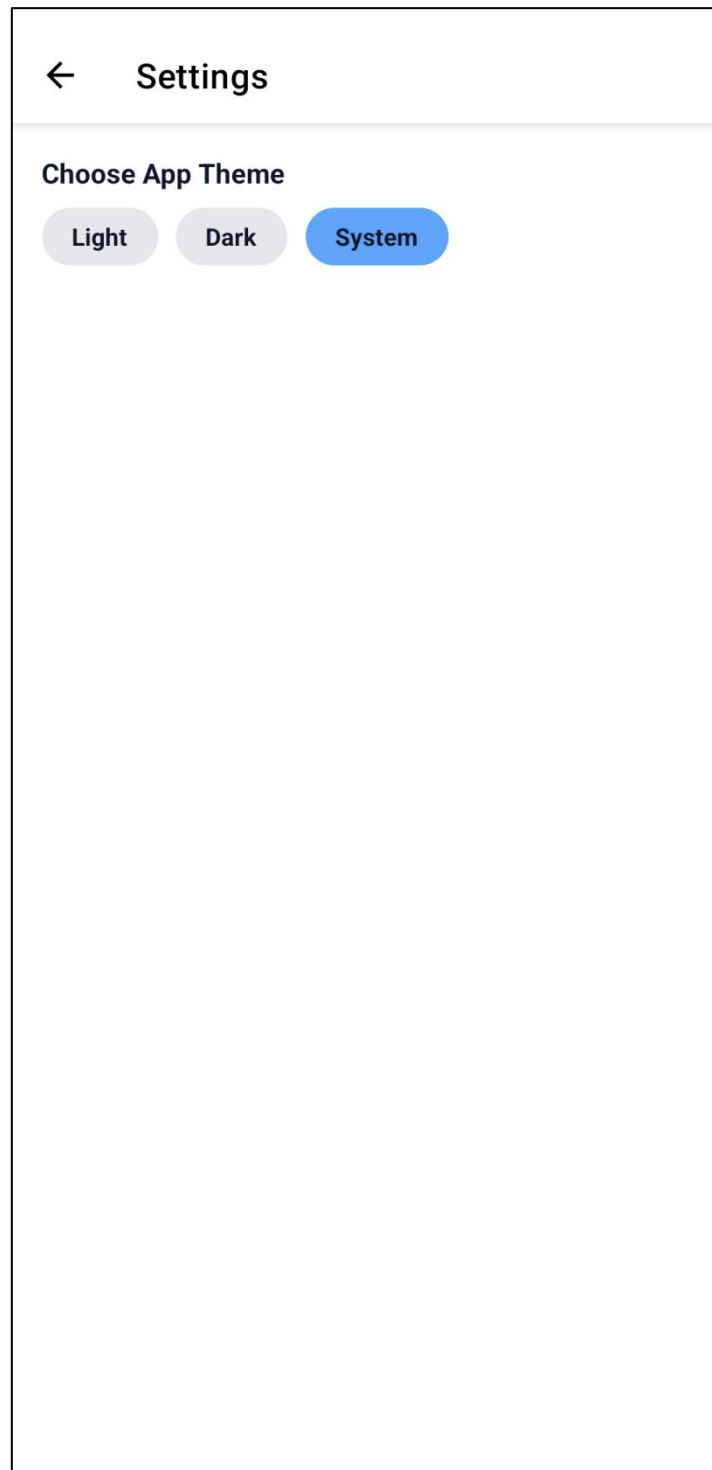


Figure 13 : Settings Screen

Settings Screen can change application theme (Dark, Light and System)

New Chat Screen

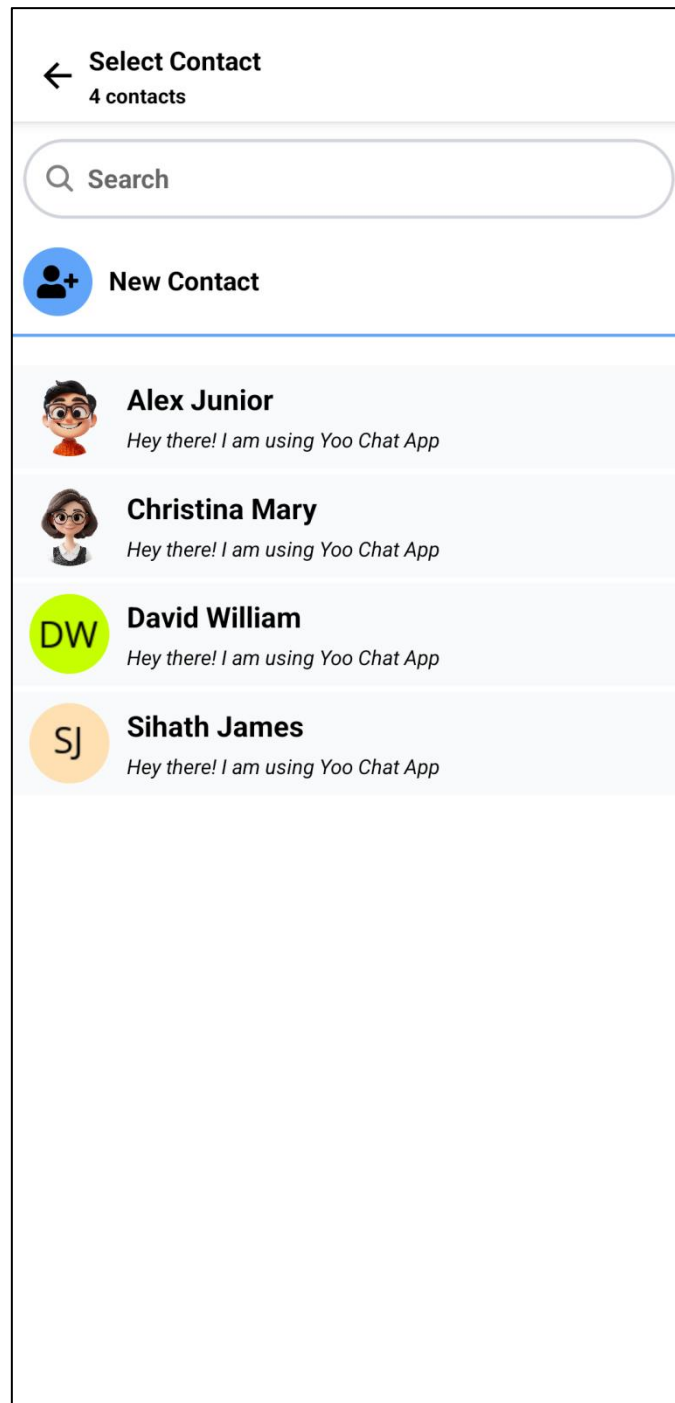


Figure 14 : New Chat Screen

This screen show user's friends. User can touch a friend and start new chat. User can search friends. Also there is a new contact button for create new contact.

New Contact Screen

 **New Contact**



First Name



Last Name

 Sri Lanka ▼



+94

Phone

Save Contact

Figure 15 : New Contact screen

This screen enable to save new contact for user.

Status Screen

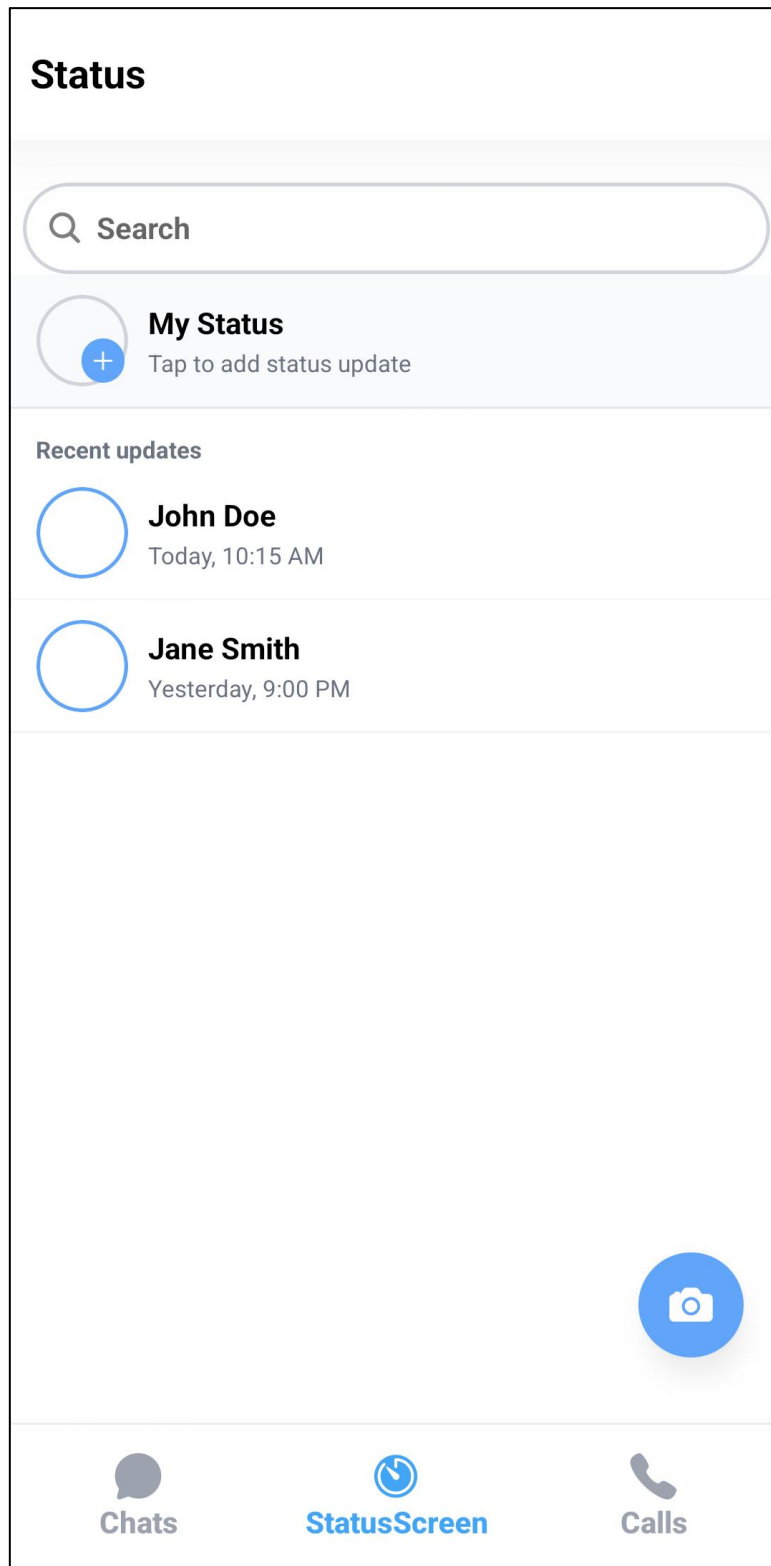


Figure 16 : Status Screen

Calls Screen

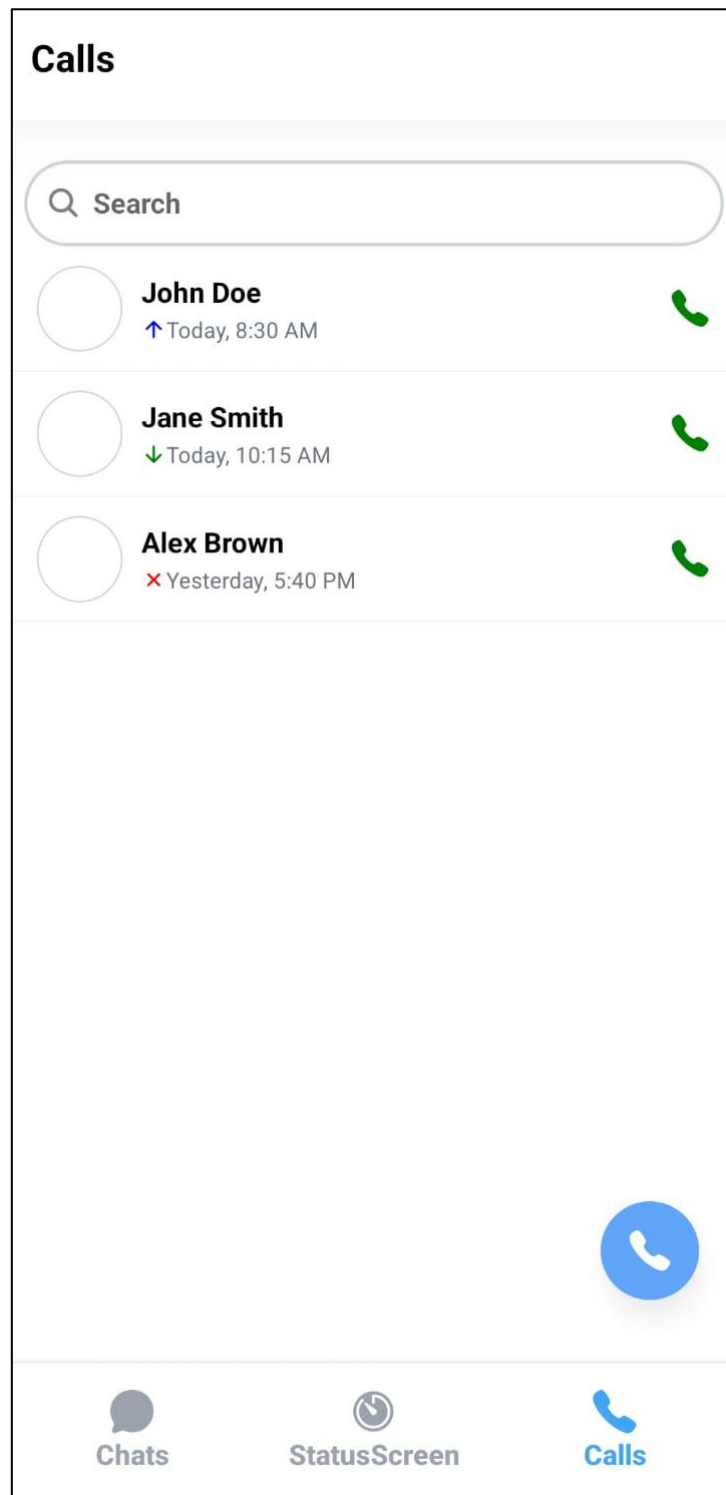


Figure 17 : Calls Screen

Diagrams

Usw Case Diagram

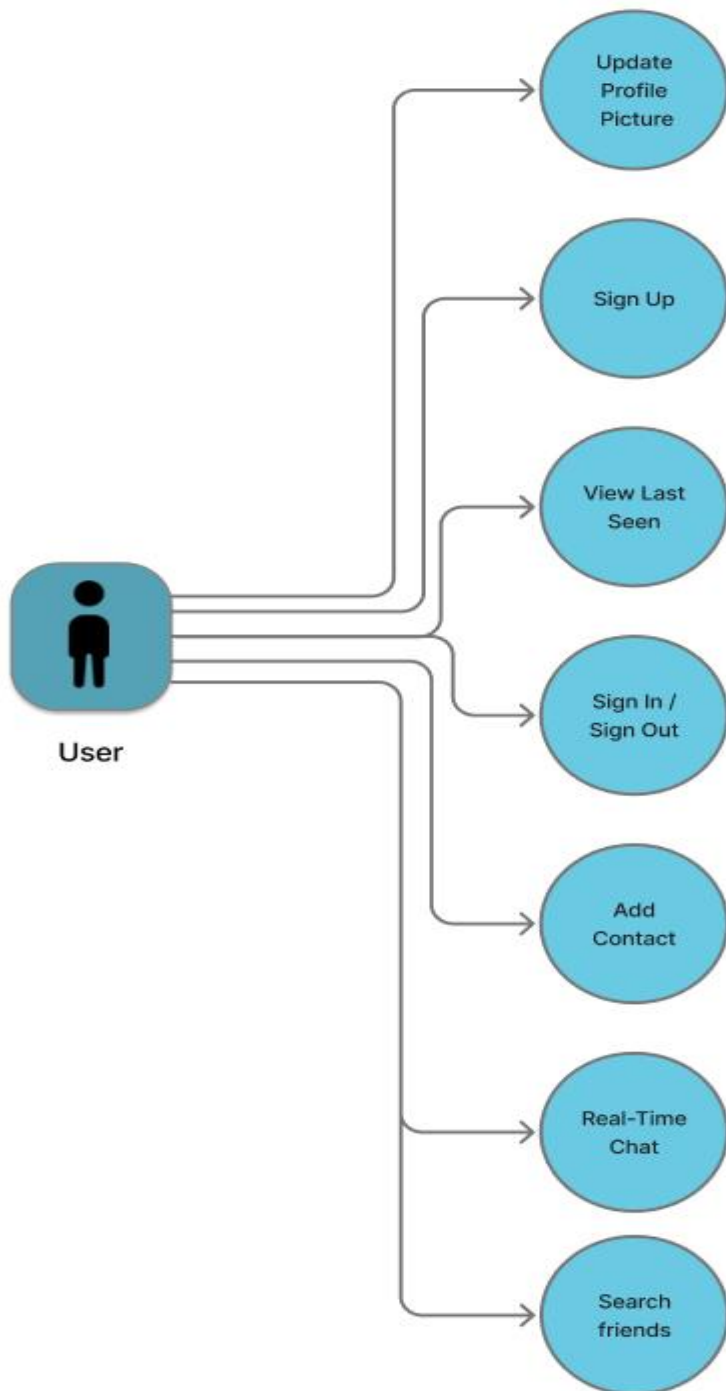


Figure 18: Use Case Diagram

Class Diagram

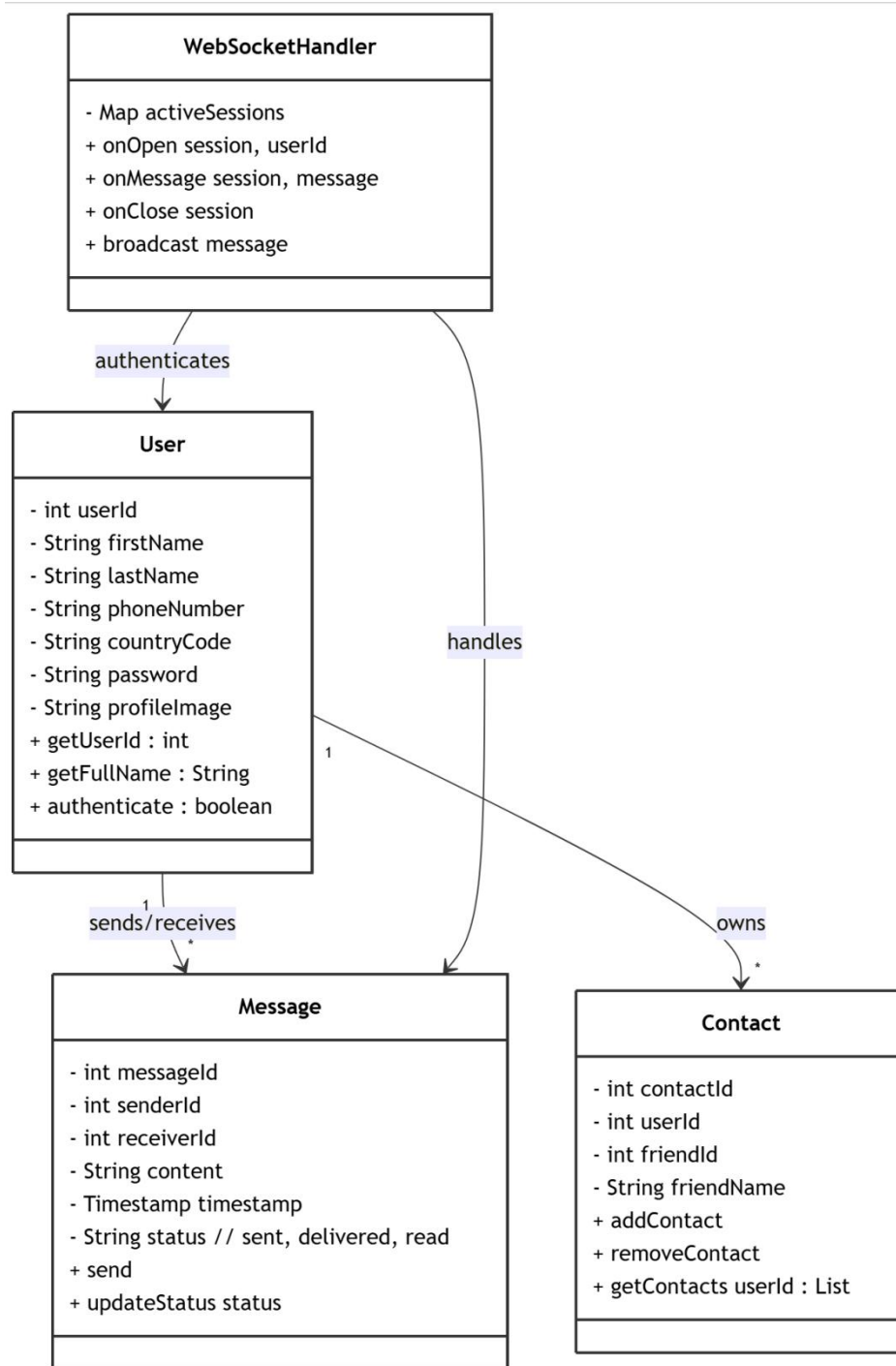


Figure 19: Class Diagram

ER Diagram

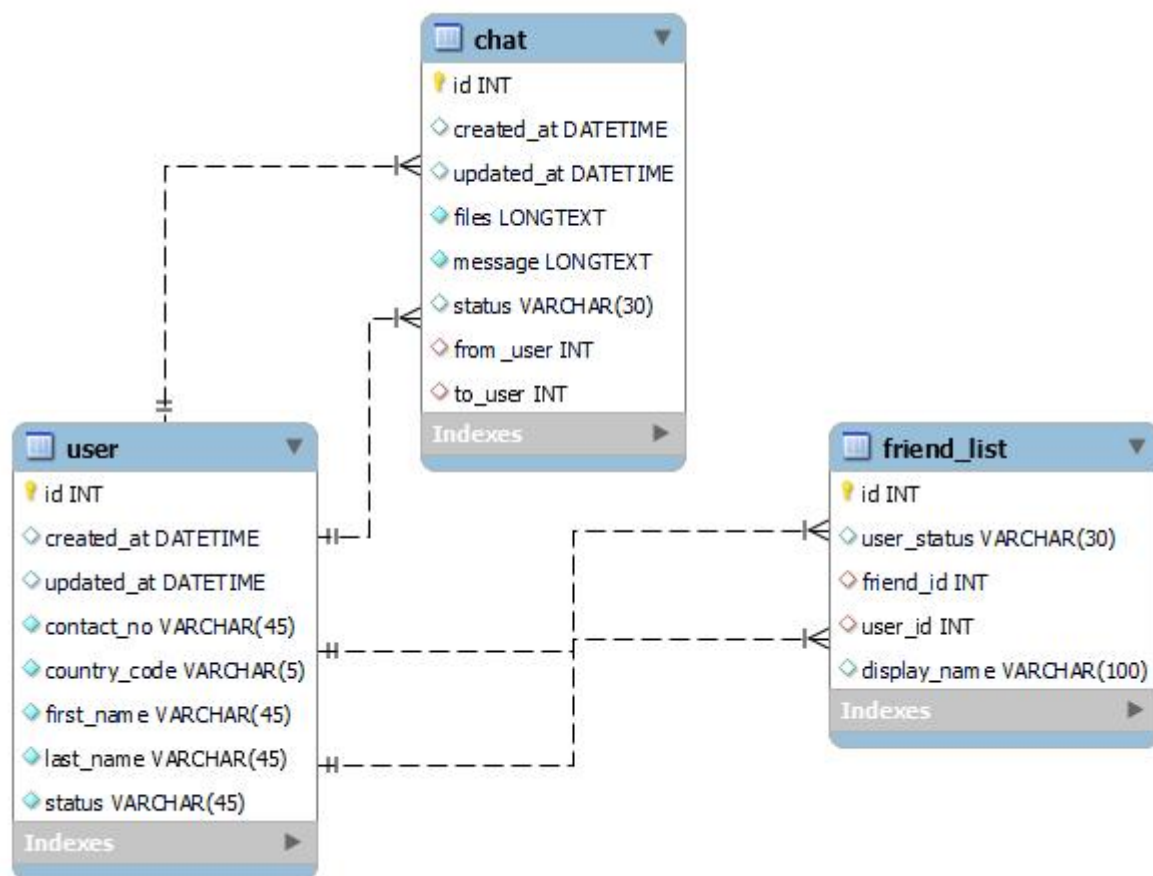


Figure 20: ER Diagram

Sequence Diagram

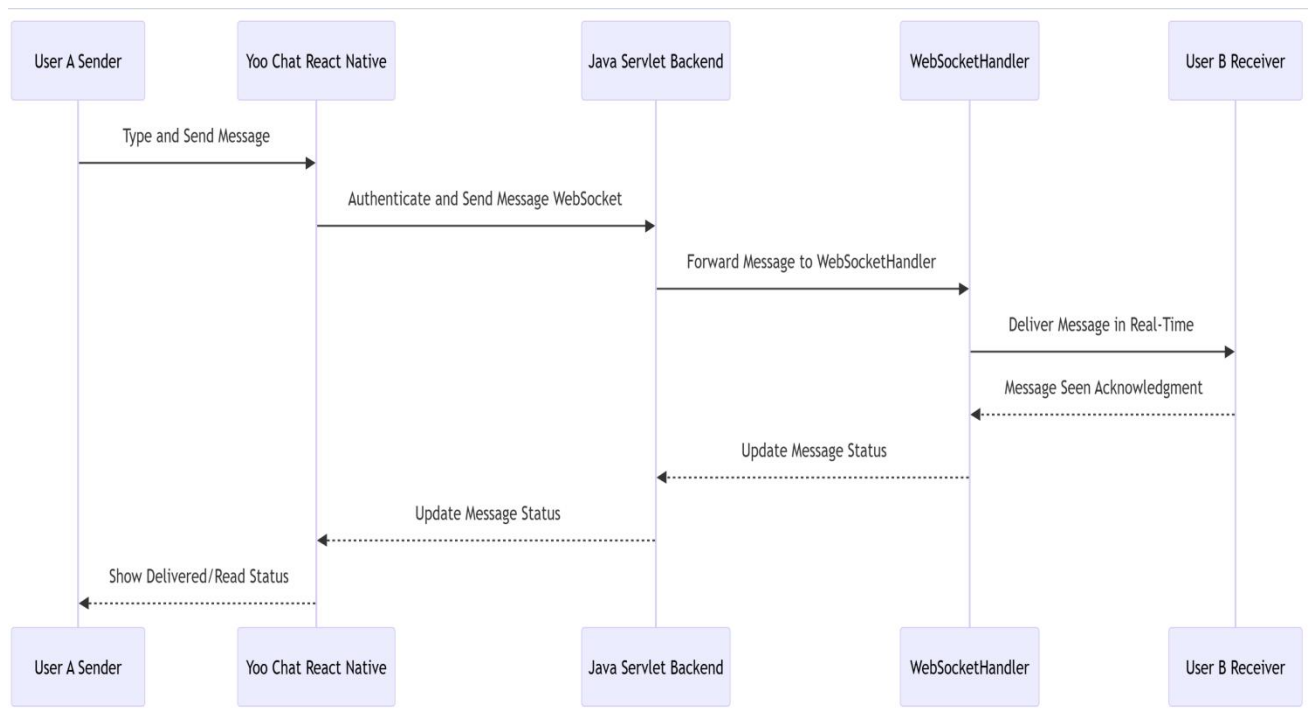


Figure 21: Sequence Diagram

Backend Implementation

- User Authentication and Registration

User sign up requests handled through Java Servlet APIs, that validated user details and manage session information. User details includes user, county code contact number, profile image, User's first name and full name.

- Database Interaction

Hibernate ORM used to map java objects to relational tables in the MySQL database. Tables:User,chat and friend list. It reduced SQL codes.

- Real time messaging

The Yoo Chat App used Websocket connection to ensure real time communication between client and server. When user send message to someone, the message is received by the server's WebsocketHandler and instantly send message to the right person without screen refreshing.

- REST APIs for data fetching

The backend uses the REST APIs to handle data retrieval between client side and server side.

REST endpoint are responsible for:

- ◆ Fetching user information
- ◆ Loading chat history
- ◆ Contact management
- ◆ Profile data retrieval

All responses are returned is JSON format. While WebSocket managing real time messages, the REST APIs ensure persistent data.

Frontend Implementation

The Yoo Chat app frontend is build bu using React Native and it ensures a smooth responsive user experience both Android and iOS.

- React Native Components of UI

This app used custom components such as chat bubbles, friend list, buttons, options and input fields to give modern and attractive UI for the users. Also FlatList are used for the long lists like chat list, thouchableOpacity and other compornents are provide smooth navigation and actions.

- State Management

The Yoo Chat App used React Hooks such as useState and useEffect to update chat list, messages and user data in real time. That ensures the UI updating dynamically with the backend data without screen refreshing.

- **WebSocket Integration**

The Websocket connection established to enable real time data transferring between users. Incoming messages status are handling via event listeners with immediate UI updates for best chat experience.

- **Animations and alerts**

For the smooth transitions between screen, this app used smooth animations. The toast messages are used to notify errors, warnings and success messages.

Database Design

Tables:

user : id, created_at, updated_at, contact_no, country_code, first_name, last_name, status

chat : id, created_at, updated_at, files, messages, status, from_user, to_user

friend_list : id, user_status, friend_id, user_id, display_name

Security Measures

To ensure secure user experience, the Yoo Chat App implements multiple security layers

- **Authentication and session management**

User authentications are handle on REST APIs while sign up and load user. When load app it retrieves the user data such as User Id and restore the user session. The loading state (isLoading) ensures that UI is waits until complete session validation. Sensitive identifiers are stored securely using AsyncStorage.

- **Securing WebSocket Connection**

Used ping pong technology for the WebSocket connection continuity. Each websocket provide includes authenticated user id.

- **Data Privacy and Access Control**

Users can only access their chats by involving them. Also this app used input validations in both frontend and backend, and it is preventing unauthorized access and data modifications.

Testing

- **Unit Testing** : This app used individual components testings, such as message sending function, message status updating function, profile picture updating function, new contact function and listing components.
- **Integration Testing** : Tested to ensure data flow between frontend and backend.
- **Real time testing** : Tested real time messaging and status changing by using two mobiles.

Deployment

The Yoo Chat App deployed using combination of ngrok and expo go app to ensure itegration between frontend and backend.

Backend Deployment : The backend is deployed on the GlassFish server to handle java servlets, WebSockets connections and APIs. The database connected on locally through hibernate ORM. This app used ngrok to expose the app for mobiles through the internet.

Frontend : The front end of the Yoo Chat application tested on Expo Go app.

Future Enhancements

- Send real time notifications for new messages.
- Media and file sharing.
- Implement message end to end encryption for the better privacy.
- Group chat creation.
- Add typing indicators.

Conclusion

The Yoo Chat App shows the itegration between frontend and back end components. Frontend is build in React Native and the backend is biuld in Java Servlet. The backend functions are running on GlasFish server. The Yoo Chat App provides real time chat feature using WebSocket connection and APIs for data management.

The application provides secure authentication and efficient data handling for User friendly and smooth interface interactions. The project reaches both functional and reliable status through its well designed backend services and database connections and frontend state control system.

The project demonstrates how React Native and WebSocket and Java EE technologies work together to create a scalable interactive secure chat system. The application will reach full production readiness through future development of media sharing capabilities and encryption methods and notification systems.